

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class: TYIT

Sem: VI

Roll No.: TYIT29

Date :

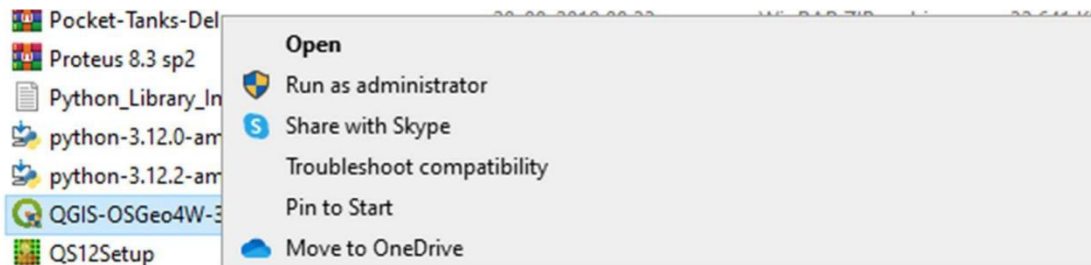
Course Name : Geographical Information System

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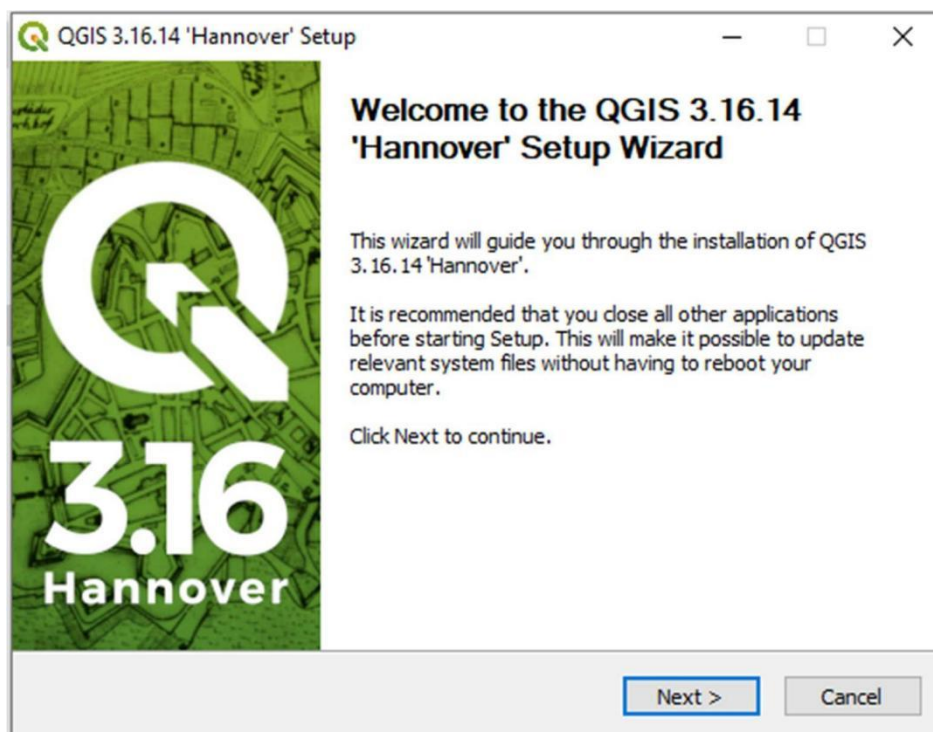
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Q:- 1 Familiarizing Quantum GIS: Installation of QGIS, datasets for both Vector and Raster data, Maps.

Step 1:- Find the file and select **Run as administrator**.



Step 2:- Click on Next.



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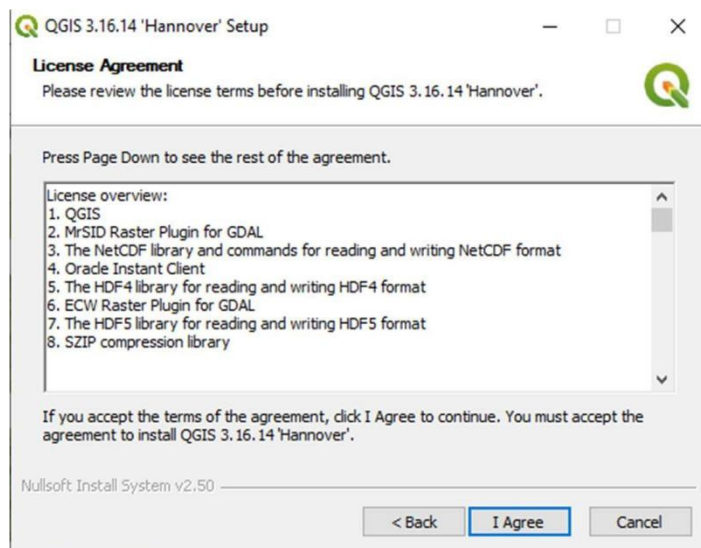
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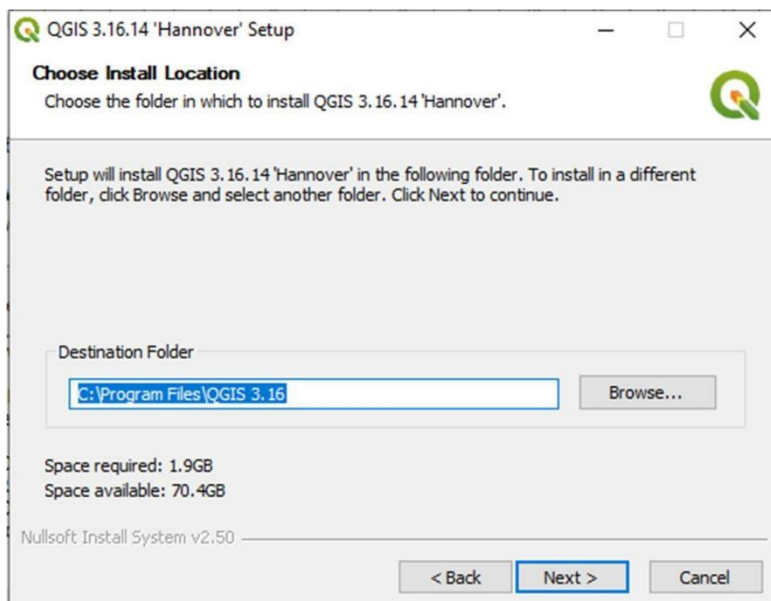
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Step 3:- Agree the License Agreement by clicking on **I Agree**.



Step 4:- Select the Installation Location and click on **Next**.



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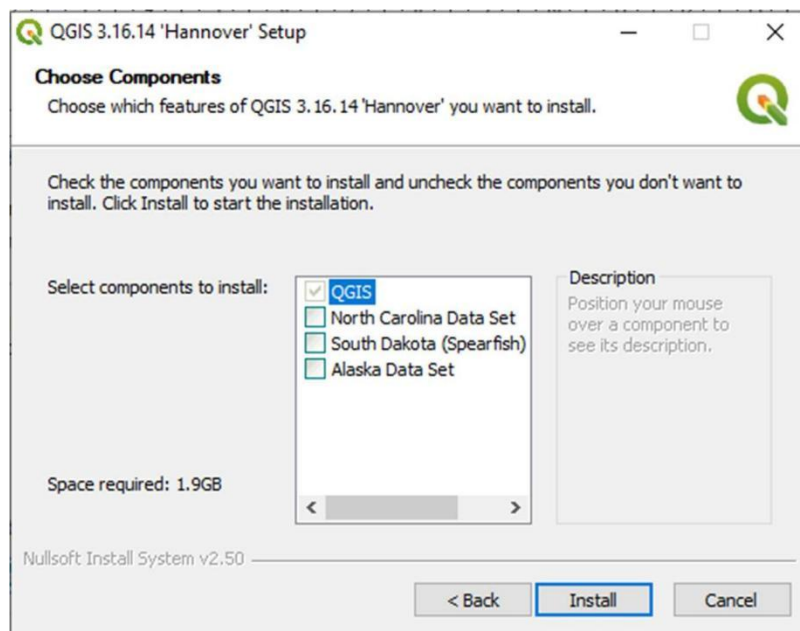
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Step 5:- Select the QGIS and Install it by clicking on **Install**.



Step 6:- Wait for unzipping the file and then click on **Finish**.



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Creating and Managing Vector Data:

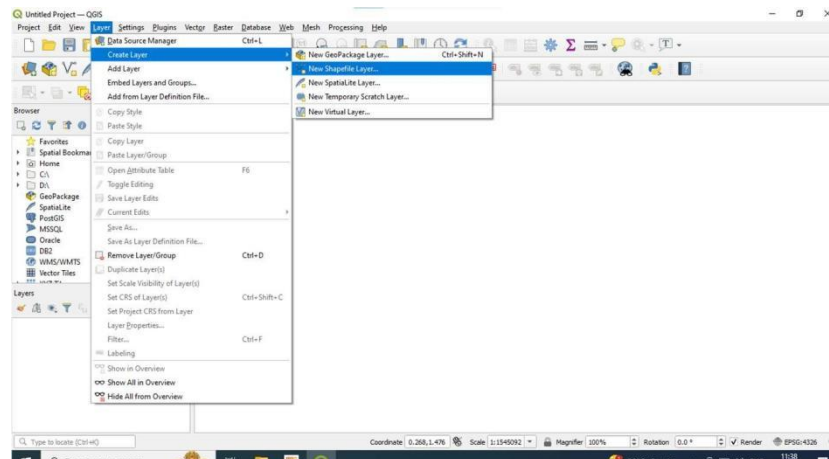
- a) Adding vector layers
- b) setting properties,
- c) formatting

Step 1- Create a New Project on QGIS

- Open QGIS.
- Click on "**Project**" in the top menu bar.
- Select "**New**" to start a new project.

Step 2- Create a New Shapefile Layer

- Click on "**Layer**" in the top menu bar.
- Select "**Create Layer**" from the dropdown menu.
- Choose "**New Shapefile Layer**" from the options.



Step 3- Name and Save the Shapefile

- In the "**File Name**" field, enter a name for your shapefile (e.g., andheri.shp).
- Click the three dots ("...") next to the "**File Name**" field to choose a folder path where the file will be saved.
- Set the **Geometry Type** to **Polygon**.
- In the "**New Field**" section, type the name of the field you want to add (e.g., id or name).
- Specify the data type for each field (e.g., Integer for id and Text for name).
- Click on the "**Add to Fields List**" button to add each field.
- After adding the desired fields, click "**OK**" to create the shapefile.

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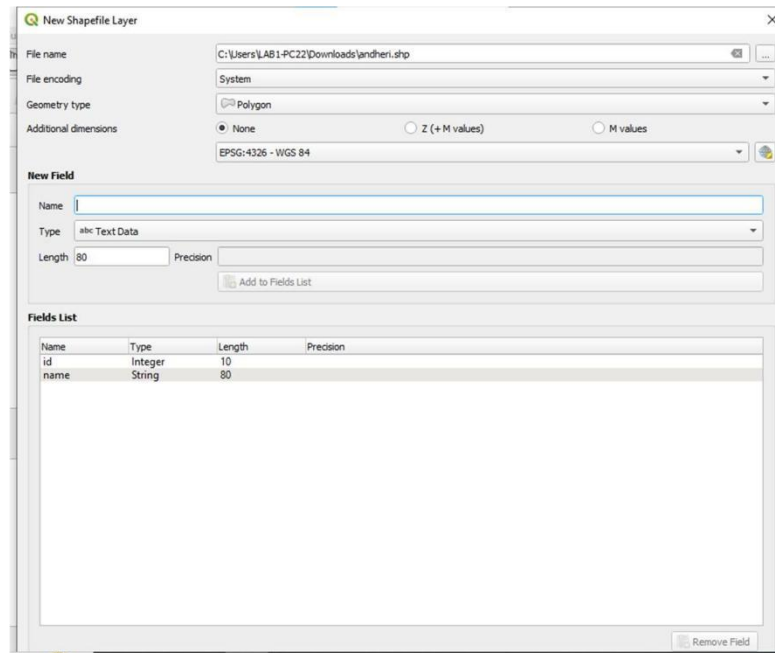
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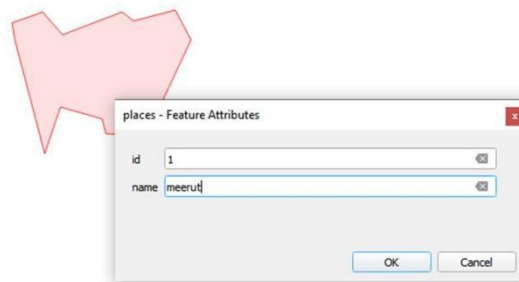
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Step 4- Create the First Polygon

- Start editing the layer by clicking on the **"Toggle Editing"** button in the toolbar.
- Use the **"Add Polygon Feature"** tool to draw your first polygon on the map.
- A pop-up dialog will appear prompting you to input values for the fields you defined earlier. Enter the id (e.g., 1) and the name (e.g., Andheri).
- Click **"OK"** to save the polygon with the specified attributes.



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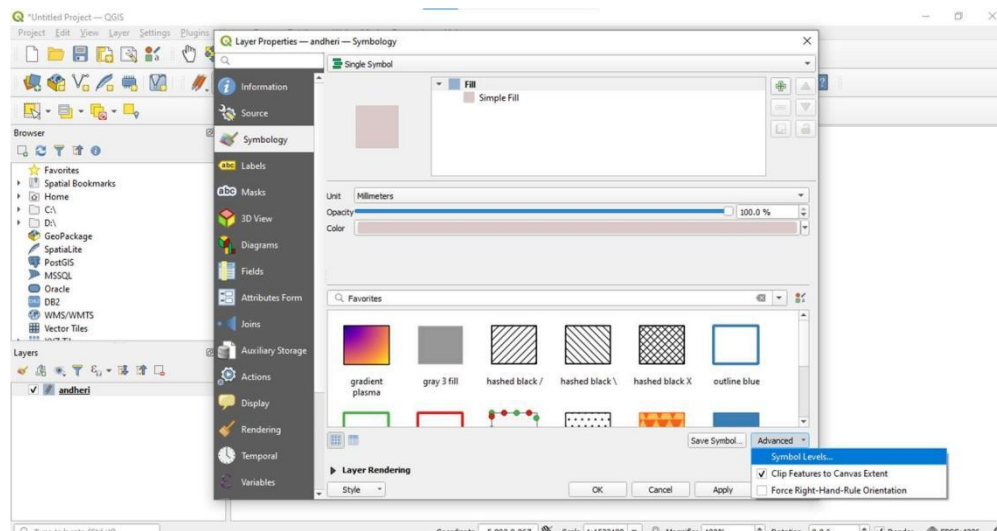
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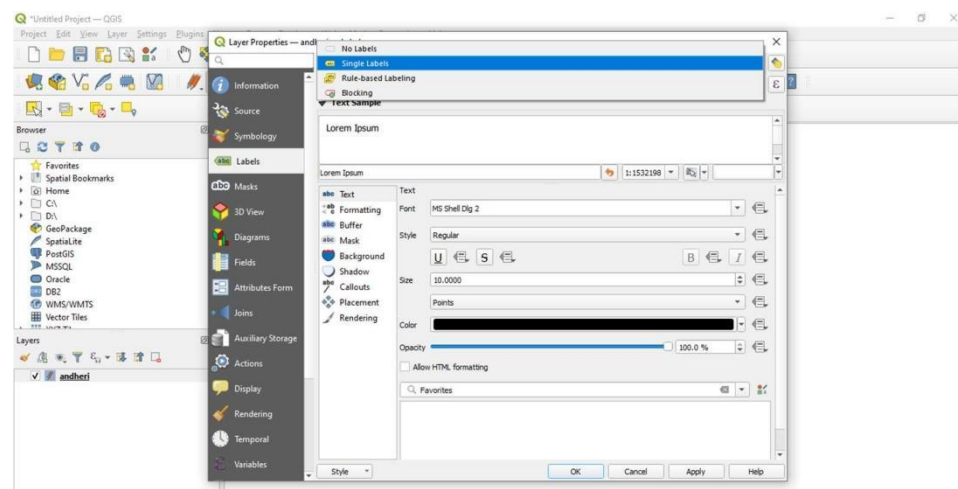
Step 5- Change the Fill Color

- In the "**Layer Properties**" window, go to the "**Symbology**" tab.
- Under "**Symbol Layer Type**", choose "**Simple Fill**" (if not already selected).
- Click on the "**Fill Color**" box to select a new color for your polygon.
- Adjust the **Stroke Color** and **Stroke Width** if needed.
- Click "**OK**" to apply the changes.



Step 6 – Enable Labels

- In the "**Layer Properties**" window, click on the "**Labels**" tab in the left-hand menu.
- Under the "**Label With**" drop-down menu, select "**Single Labels**"
- Click "**OK**" to apply the changes.



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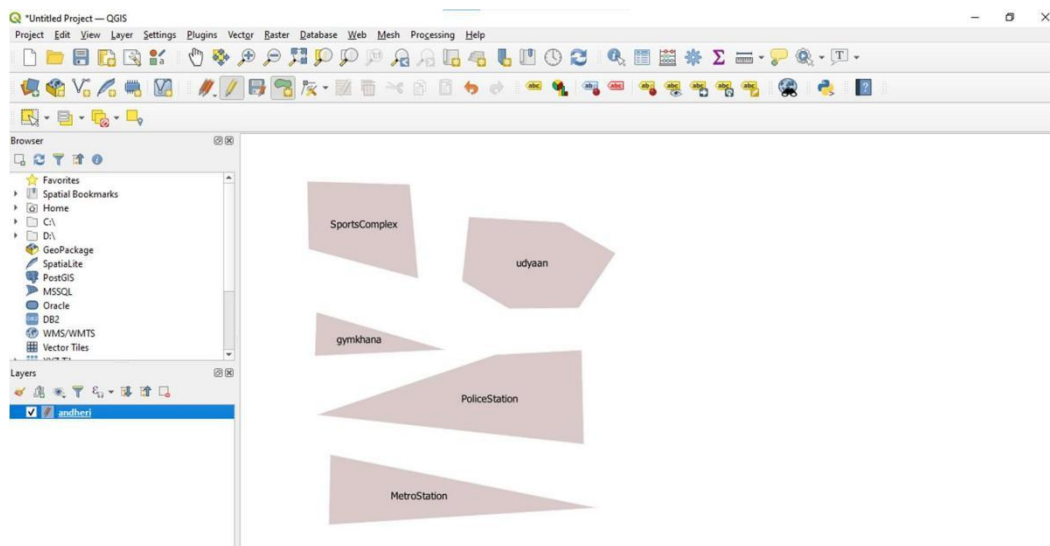
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You can now successfully create and customize polygons



Now , like polygons u can Create Lines Defining Roads.

Step 7 - Create a New Shapefile Layer for Lines

- Click on "**Layer**" in the top menu bar.
- Select "**Create Layer**" and then "**New Shapefile Layer**".
- In the "**File Name**" field, name the layer (e.g., roads.shp).
- Set the Geometry Type to **Line**.
- In the "**New Field**" section, specify the fields you want to add (e.g., id for road ID, name for road name).
- Click "**Add to Fields List**" and then "**OK**".
- After drawing each line, a pop-up dialog will appear asking you to enter values for the fields (e.g., id, name).
- Enter the values and click "**OK**" to save the line.

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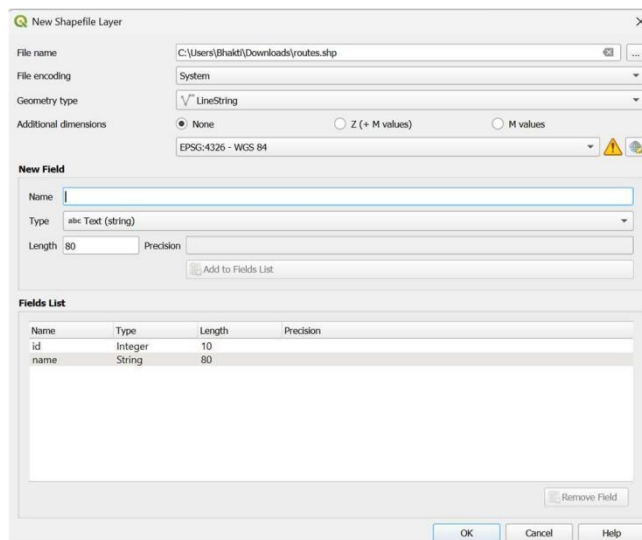
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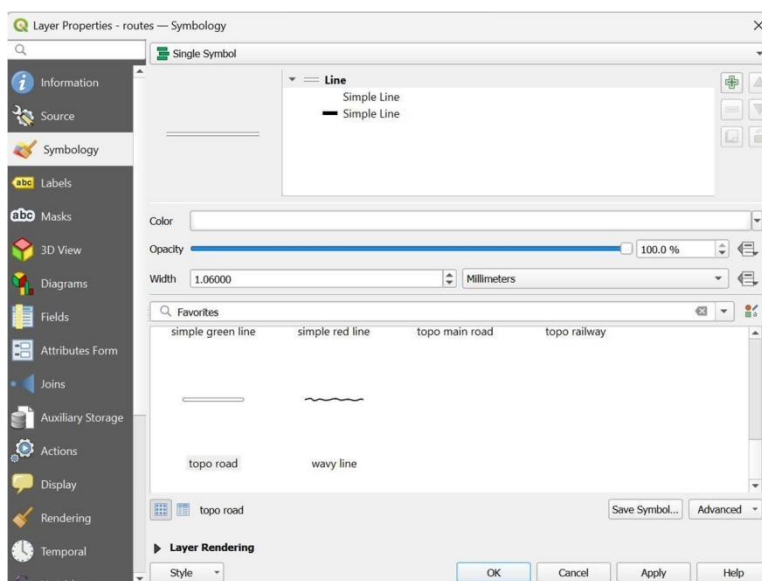
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Step 8 - Customize Line Appearance

- Right-click on the roads layer in the **Layers Panel** and select **"Properties"**.
- In the **"Symbology"** tab, click on the drop-down menu for **"Symbol"** and select **"Topographic Roads"**.
- The "Topo Road" symbol will apply, giving your roads a standard visual representation used for road layers.
- Adjust other properties like **Line Width** or **Color** if desired. Click **"OK"** to apply the change.



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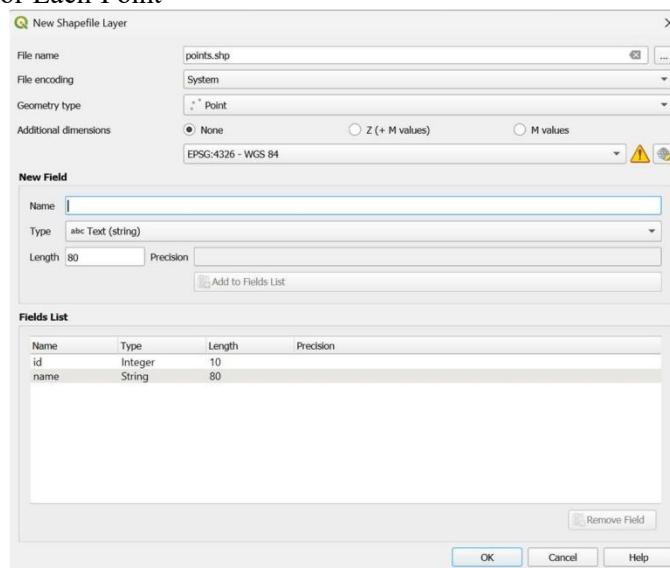
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Steps to Create Points in Your Project

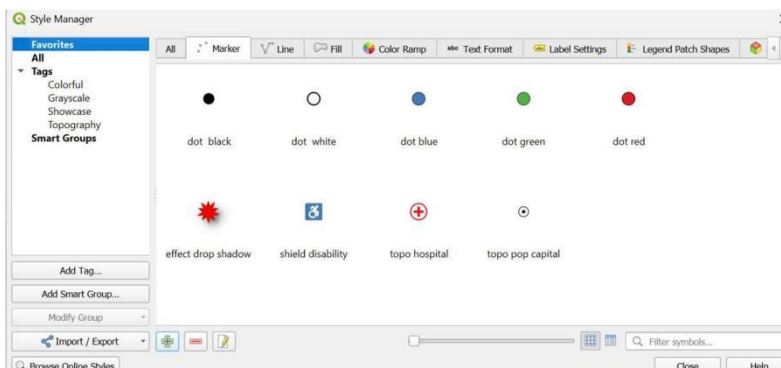
Step 9 - Create a New Shapefile Layer for Points

- Click on "**Layer**" in the top menu bar.
- Select "**Create Layer**" and then "**New Shapefile Layer**".
- In the "**File Name**" field, name the layer (e.g., points.shp).
- Set the **Geometry Type** to **Point**.
- Add Fields to the Point Layer
- Input Attributes for Each Point



Step 10 - Change Point Symbol

- Right-click on the points layer in the **Layers Panel** and select "**Properties**".
- In the "**Symbolology**" tab, click on the "**Symbol**" drop-down menu.
- Select a point symbol (e.g., a circle, square, or custom image) to represent your points.
- **Adjust Point Size and Color**
- Click "**OK**" to apply the changes and display the points with your customized symbol.



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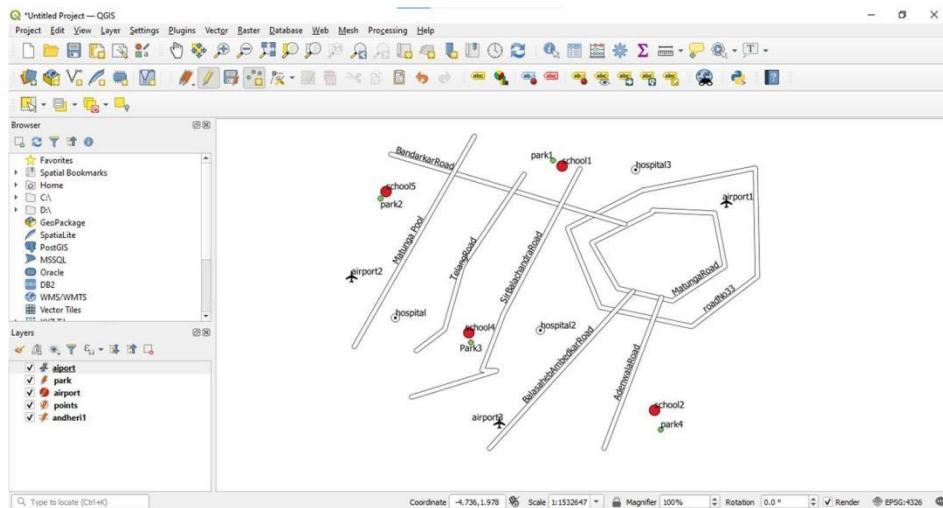
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You can now successfully create and customize lines and points on your project



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